DEEPAK R

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EDUCATION

Dhaanish Ahmed College of Engineering (Anna University) *May 2024 – May 2028*
Bachelor of Engineering (B.E.) in Robotics and Automation ***(CGPA: 9.2 / 10.0)*** *Chennai, India*

TECHNICAL SKILLS

Languages: Python, C++, C, Rust, TypeScript, Bash, SQL
Robotics & Simulation: ROS 2 (Jazzy), MoveIt 2, MuJoCo, LeRobot, Gazebo, Nav2, SLAM Toolbox, SUMO / TraCI
AI & Computer Vision: PyTorch, Diffusion Transformers (DiT), Flow Matching, Model Context Protocol (MCP), OpenCV, YOLO
Embedded & Systems: STM32, ESP32 (FreeRTOS), Raspberry Pi 4, BTS7960 Drivers, Linux (Ubuntu), Git, Fusion 360

EXPERIENCE

Wildplant Terrestrial Solutions Pvt Ltd *Jul 2026 – Present*
Frontend & 3D Web Development Intern (NDA Protected) *Remote*
- Engineered responsive user interfaces and modular 3D viewport modules using Three.js, React, and WebGL.
- Optimized client-side rendering pipeline and state management for proprietary industrial platforms.

Precise3DM *Jun 2026 – Jul 2026*
AI & Data Engineering Intern *Chennai, India*
- Architected automated B2B client qualification pipeline using Playwright and Google Gemini API.
- Constructed automated data sanitization workflows to extract, validate, and export structured datasets.

Tamizhan Skills (RISE Programs) *Dec 2025 – Mar 2026*
AI & Autonomous Systems Intern *Remote*
- Configured ROS 2 Nav2 navigation stacks, costmaps, and multi-sensor fusion (LiDAR, IMU, Odometry) in Gazebo.
- Trained CNNs in TensorFlow/Keras for image classification and developed NLP anomaly detection pipelines.

PROJECTS & RESEARCH

VGA Policy: Vision-Geometry-Action Robotic Manipulation *Aug 2026 – Oct 2026*
PyTorch • LeRobot • MuJoCo • Flow Matching DiT *LIBERO-Spatial Benchmark*
- Developed ≤0.5B parameter (298M) embodied policy achieving **154–215 Hz** hard real-time control rate (4.6ms motor latency).
- Physics-guided kinematic loss annealing reduced jerk by **89.1%**; verified closed-loop pick-and-place success in MuJoCo.

NeuroCode: Heterogeneous Code-Graph Slicing *Oct 2026 – Nov 2026*
Python • MCP Server • Tree-sitter • NetworkX *Published on PyPI: neurocode-ai*
- Authored neuro-symbolic MCP agent resolving SWE-bench Lite bugs on 4GB VRAM laptop GPUs (961MB peak) with 0 OOMs.
- Achieved 100% recall on RepoBench-R with 42.6% token reduction; prepared submissions for IEEE TSE and Elsevier JSS.

Orbital ADR: Autonomous Space Debris Cleaner *Sep 2026*
ROS 2 • Jetson Nano • STM32 • Fusion 360 *Smart India Hackathon 2026 (PS SIH26226)*
- Engineered LEO active debris removal spacecraft linking Jetson Nano SpaceYOLO pose estimation with STM32 cold-gas thrusters.
- Combined 4-piston kinetic damping with electroadhesive capture plates to secure non-cooperative tumbling space debris.

AURA: Semi-Markov RL Smart Traffic Management *Aug 2026 – Nov 2026*
PyTorch • SUMO • TraCI • React • Node.js DDS *Zenodo DOI: 10.5281/zenodo.20265628*
- Architected SMDP DQN with rolling emergency vehicle preemption (EVP) evaluated on 177-intersection Manhattan network.

PUBLICATIONS & HONORS

Preprint: *Sensor-Fusion Driven DRL for Dynamic Traffic Signal Optimization* — Zenodo DOI: 10.5281/zenodo.20265628 (2026)
Kaggle Global Rank #72 (Top 100): Kaggle "Name That LLM" Competition (Score: 0.639) with Dual 14B Qwen LoRAs.
ARC Prize 2026 Contender: NeuroARC spatial-graph slicing agent achieving 510 FPS and 98.4% entity interaction hit rate.
Hackathon Finalist: Codex 48 Finalist (Arkz AegisBrand); Smart India Hackathon 2026 Hardware Space Tech.